Wide View and Line Filter for Enhanced Gradient Computation in Aquatic Environment

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***Abstract*— In this paper, two new gradient operators, the wide view filter (WVF) and line filter (LF) based on a two-dimensional high-order Taylor Series expansion are proposed to tackle problems with gradient computation in aquatic environment. Different from most existing kernels, the WVF evaluates a wide range of pixels (limited only by image size and computational requirements) to effectively suppress noises and offer a more accurate estimation of numerical derivatives and gradients in low signal to noise (SNR) images. Leveraging strengths of WVF, the LF operator combats the next challenge associated with degraded and low contrast line boundaries. Its novelty lies in aggregating the gradients computed by multiple WVFs along a specific line orientation, resulting in smooth edge detection in the presence of significant discontinuities, obstructions, poor edge quality, and low contrast.**

Keywords—Image Processing, Gradient Computation, 2D Filtering, Taylor Expansion, Edge Detection, Noise

I. Introduction

Pixel gradient calculation is a challenging yet vital step in virtually all image processing pipelines. Its accuracy directly affects the success of many subsequent processing techniques including segmentation, feature extraction, edge detection, object detection, among others, all of which are critical in autonomous vessels. Challenges for accurate gradient computation in both magnitude and orientation arise from various factors, such as signal noises, varying light intensities, blurriness, and directional decomposition [1]. These challenges are especially detrimental to image processing techniques in underwater and littoral environments where lack of light, color attenuation, water clarity, adverse weather, reflectance, refraction, and small objects more significantly impact image quality and the ability to produce accurate image gradients. Traditionally, image gradients have been produced using small filters leveraging the finite difference method. While easy to apply, they are highly susceptible to noise and suffer from low contrast and disconnected edges [2]. Smoothing filters are commonly used to address image noise, but these degrade line boundaries, smooth valuable textural information, and blur small objects. Recently, deep learning methods have taken over, because they display high accuracy with a relatively low computational load. However, they still suffer from challenges like data scarcity, generalization to various environments, and even traditional problems like noise. [3]

II. Related Works

Many gradient detection filters are reported, each with their own merits and disadvantages. Most common gradient-based operators are Sobel [4], Prewitt, and Roberts kernels which are typically 3x3 patterns convolved around an image to produce the intensity derivatives of each pixel [5]. Occasionally, larger variations in sizes 5x5, 7x7, and more are used to suppress noise in images and improve the resultant features [6]. While easy to apply to an image, their accuracies depend on image quality. The simplistic structure of these kernels limits their performance, and their high pass nature amplifies noise [2] [1]. Smoothing filters are commonly used to reduce noises but deteriorate image quality and make subtle gradients undetectable either. External factors like refraction, poor focus, object obstruction, low light, and smooth intensity transitions can also affect the quality of gradient calculation.

Alternative methods to gradient-based operators for edge detection that delve into second derivative zero crossings, region segmentation, image morphology, fuzzy logic, spatial frequency, machine learning, and more have been proposed to address their common issues [3]. The Laplacian of Gaussian [7] filter was proposed as an alternative to single differentiation with better orientation independence; however, it displays extreme sensitivity to noise and inherently degrades the image information using a Gaussian smoothing filter [8]. Region based [8] methods as Khan proposed [9] are very accurate in describing high level edges but lack all textural information resulting in an oversimplified edge map. Before deep neural networks gained popularity, frequency domain analysis that locates features at maximum phase congruency was considered amongst the best. [10] Fuzzy logic edge detectors have been developed based on the principle of mathematical and logical reasoning to reduce complexity. It, however, still requires gradients as it essentially classifies edge pixels from gradient profiles. In addition, these methods are typically sensitive to noise unless paired with a high order manually determined ruleset. [11] Lin proposed a new high-pass filter [1] for determining edges in classically noisy medical images that does

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improve response in unideal noise conditions. It still does lack angular sensitivity, and suffers from noises, and low contrast/degraded boundaries.

Machine learning-based edge detection models have displayed a high degree of accuracy with relatively low computational load. [12] DexiNed [13], BDCN [14], EDTER [15], and TEED [16] are popular edge detection models that have shown great promise, and top the quantitative metrics for edge detection online. Though impressive, they are not immune to the common disadvantages of machine learning. On the other hand, machine learning-based methods require a vast amount of data for model training. This presents an inherent problem with model generalization as it is restricted to finite data and cannot interpret situations which it was not specifically trained on. [16] In niche settings, like littoral environments, datasets are extremely scarce, and the environment is highly cluttered and diversified. Deep learning models have difficulty replicating the human vision system in these scenarios because they lack certain intangible human characteristics such as common sense, intuition, explanation, and the ability to process foreign situations. Gradient-based operators are not subject to the constraints of data as they define features according to their physical representation. They are also progressing to mitigate computational burden on fieldable systems and ensure reliability in handling challenges associated with poor image quality, such as extreme SNR. A systematic approach that can directly operate on the original image's raw pixel intensities, accurately calculate pixel gradients, and address abovementioned issues with minimal manual manipulation is still needed.

III. Overview Of Solution

Calculating pixel gradients requires a highly robust derivative or gradient operator to address challenges associated with low SNR, inconsistent scene illumination, and broken feature contours or obstructions. Accurate gradient orientation and magnitude determination must resolve these issues. Gradient operators resemble image sharpening filters, which exacerbate high frequency image components, posing a problem in obtaining spatial derivatives. In this context, a new Wide View Filter (WVF) is proposed in the present study, which incorporates significantly more image information compared to other existing kernels [17]. This is consistent with the human vision system which is generally unaffected by noise in its wide view [2]. The WVF calculates normal and tangential derivatives along a specified orientation (θ) at a target pixel by treating the values of surrounding pixels as a high-order Taylor series expansion. Using many surrounding pixels (hence, termed wide view) and a high-order Taylor expansion suppresses noises in a statistical manner, achieving an unprecedented accuracy of gradient computation.

Yet, the WVF is still less effective in handling low contrast and broken boundaries and weak gradients along an edge, caused by varying image contents and light intensities across the spatial domain. To address this issue, a new Line Filter (LF) has been developed. The LF defines a line centered around an origin (target) pixel and determines its numerical derivatives using weighted derivative values of multiple auxiliary virtual pixels along the established line oriented in a specified direction. These derivatives are calculated with the WVF. When a faint edge aligns with the LF orientation, the gradient in its normal direction will be amplified significantly as it aggregates the values of all its neighbor pixels in the edge direction. Thus, the LF compensates low-gradient areas using gradient information in the neighborhood along the line, enabling smooth edge detection in the presence of extreme noise and obstructions and precise determination of edge orientation.

A. Wide View Filter Architecture

The WVF is based on a two-dimensional Taylor Series expansion and calculates high-order derivates at the target pixel using a least-squares approach. It employs a unique structure and differs from common gradient operators such as Sobel, Gaussian, and others that are square kernels centered around a target pixel to calculate the horizontal and vertical gradients, which are then combined to calculate the magnitude. Therefore, they are susceptible to noises and inaccurate for edge orientation determination. The proposed WVF, on the other hand, mitigates this issue by incorporating a number (N_p) of pixels in a circular region (see the light-yellow region) around a target (origin) pixel to calculate the normal and tangential derivatives along a specific orientation θ, as shown in Figure 1, where N_p is significantly larger than common operators and improves its robustness to signal noise. N_p is only limited to the size of the image and any computational constraints.

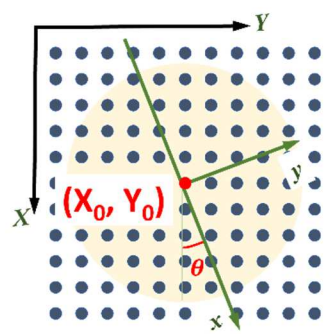

Figure 1: Wide View Filter Schematic and its Global and Local Coordinate System

The nearest neighbors of the origin pixel define a local coordinate system denoted as (x, y), which is contained in the global image coordinate system denoted as (X, Y) and located at an indexed pixel (X_0, Y_0) where $X = 1,2,\ldots,N_h$ and $Y = 1,2,\ldots,N_v$. N_h and N_v are the horizontal and the vertical sizes, respectively, as shown in Figure 1. The local coordinates (x, y) are along the tangential and the normal direction of the local coordinate system that is rotated by an angle $\theta \in [0 : 2\pi]$ with respect to (X, Y) and is used for computing the high-order derivatives at (X_0, Y_0). The WVF will be run at a number of angles $\theta = i\frac{2\pi}{N_s}$, where N_s is a user-defined number of orientations to resolve the circumferential direction and $i = 1, \ldots, N_s$ is the index. Explicitly evaluating derivatives along

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specified orientations rather than through vector decomposition from the horizontal and vertical directions is one of the novelties of our approach, significantly improving angular fidelity in gradient calculation. Note that in Figure 1 (X_0, Y_0) will be referenced as the origin hereafter, and the light-yellow region contains N_p neighbor pixels.

B. Wide View Filter Algorithm

The gradients for the origin pixel ($x = 0, y = 0$ in the local coordinate system and (X_0, Y_0) in the global coordinate system), will be calculated using the Taylor Series expansion in Equation (1). In this expression, $f(x, y)$ corresponds to the pixel intensity at (x, y) in the local coordinate system, and f^0 is the shorthand notation for *f(0,0)* For the sake of brevity, derivatives typically denoted in the following convention, f_{xxyy}, are instead represented $f_{x^2y^2}$.

$$f(x_i, y_i) \approx f^0 + f_x^0 x + f_y^0 y + \frac{f_{xx}^0 x^2}{2} + \frac{f_{yy}^0 y^2}{2} + f_{xy}^0 xy + \frac{f_{x^3}^0 x^3}{6} + \frac{f_{y^3}^0 y^3}{6} + \frac{f_{x^2y}^0 x^2 y}{2} + \frac{f_{xy^2}^0 xy^2}{2} + \frac{f_{x^4}^0 x^4}{24} + \frac{f_{y^4}^0 y^4}{24} + \frac{f_{x^3y}^0 x^3 y}{6} + \frac{f_{xy^3}^0 xy^3}{6} + \frac{f_{x^2y^2}^0 x^2 y^2}{4} + \frac{f_{x^5}^0 x^5}{120} + \frac{f_{y^5}^0 y^5}{120} + \frac{f_{x^4y}^0 x^4 y}{24} + \frac{f_{xy^4}^0 xy^4}{24} + \frac{f_{x^3y^2}^0 x^3 y^2}{12} + \frac{f_{x^2y^3}^0 x^2 y^3}{12} \quad (1)$$

The N_p pixels closest to (X_0, Y_0) are denoted as (X_i, Y_i) and (x_i, y_i) and $i = 1, 2, ..., N_p$. Then, we can replace (x, y) in Equation (1) with (x_i, y_i) and *f(x, y)* with $f(x_i, y_i)$, yielding N_p equations, as shown in Equation (2) that need to be solved to calculate all the derivative coefficients. Depending on the highest order in the expansion, the number of derivative coefficients will change. If it is the 4th order, there are 15 derivative coefficients. Therefore, we need at least 15 neighbor pixels, i.e., $N_p = 15$, to generate a system of equations with exact solutions. In general, we will consider more than the minimum number of pixels to minimize bias introduced by a few erroneous pixel values. Equation (2) can be cast into a concise matrix form and then solved in one shot using the least squares approach as show in Equation (3).

$$\underbrace{\begin{bmatrix} 1 & x_1 & y_1 & \cdots & \frac{1}{4}x_1^2 y_1^2 & \frac{1}{6}x_1 y_1^3 \\ 1 & x_2 & y_2 & \cdots & \frac{1}{4}x_2^2 y_2^2 & \frac{1}{6}x_2 y_2^3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{N_p} & y_{N_p} & \cdots & \frac{1}{4}x_{N_p}^2 y_{N_p}^2 & \frac{1}{6}x_{N_p} y_{N_p}^3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} f^0 \\ f_x^0 \\ f_y^0 \\ \vdots \\ f_{x^2y^2}^0 \\ f_{xy^3}^0 \end{bmatrix}}_{z} = \underbrace{\begin{bmatrix} f(x_1, y_1) \\ f(x_2, y_2) \\ \vdots \\ f\left(x_{N_p}, y_{N_p}\right) \end{bmatrix}}_{b} \quad (2)$$

$$\boldsymbol{z} = (\boldsymbol{A}^T \boldsymbol{A})^{-1} \boldsymbol{A}^T \boldsymbol{b} = \boldsymbol{A}^* \boldsymbol{b} \quad (3)$$

$\boldsymbol{A}^*$ in Equation (3) is the pseudo-inverse of matrix $\boldsymbol{A}$. It needs to be saved to apply the precomputed filter coefficients to a new image. In addition, in the present study, no cross derivatives are needed for the origin pixel, and only normal and tangential derivatives are extracted from **z** for use, which yields two derivatives with respect to *x* and *y* for each order specified in the expansion. A fourth-order Taylor Expansion will result in eight derivatives. After they are calculated in a specific orientation, the local coordinate axis can be rotated to any angle, and the process is repeated. Doing so changes the $\boldsymbol{A}$ matrix in Equation (2) as the positions of the neighbor pixels relative to the origin will change. Since the WVF is not based on vector decomposition along the global (*X, Y*) directions, it can accurately determine the derivative magnitude and orientation for edges.

Note that in Equation (2) the intensity value of the origin pixel, f^0, is estimated because it could be corrupted by noises. A variation of the WV filter is to treat f^0 as known if it is uncorrupted. Thus, in each individual equation, f^0, needs to be moved to the right-hand side and the first column of ones in matrix $\boldsymbol{A}$ needs to be deleted.

C. Line Filter Architecture

The LF can be considered a string of WV filters. Figure 2 illustrates the concept of the LF, and its important aspects are described as follows:

A user-defined number of pixels (in yellow) are indexed along a line centered on the origin pixel (in red). This origin pixel denotes the system (x, y). In most cases, the yellow pixels on the line will not fall on integer locations and are referred to as virtual pixels. The distance between virtual pixels on the line is kept constant as "1" while the orientation of the line can be arbitrary. Virtual pixels pose no problem to computing derivatives since they serve as the expansion point and their precise locations are the only variables needed for high-order Taylor expansion. The location of each virtual pixel will establish the nearest neighborhood of pixels to run the WV filter for each virtual pixel.

The WV filter calculates numerical derivatives for every virtual pixel along the line. The derivatives at the origin of the LF are calculated as the weighted average of all derivatives at the virtual pixels, and the weights can be determined according to the distance of the virtual pixels from the origin pixel, e.g., using a normal distribution centered on the origin pixel. Weights can be disregarded to improve the connectivity of a very subtle edge if desired.

Like the WVF, the LF described in Equations (1) and (2) will be applied at many different orientations, $\theta = i\frac{2\pi}{N_s}$, where N_s is a user-defined number of lines to resolve the circumferential direction and $i = 1, \ldots N_s$ is the index. It is expected that when the LF aligns with the edges in the image, the numerical derivatives will show the strongest values along the tangential direction. This demonstrates the capability to extract the exact orientation of an edge in an image. As a result, LF is more capable of predicting angular information compared to other filters as it contains more derivatives of an ensemble of pixels over the WV filter.

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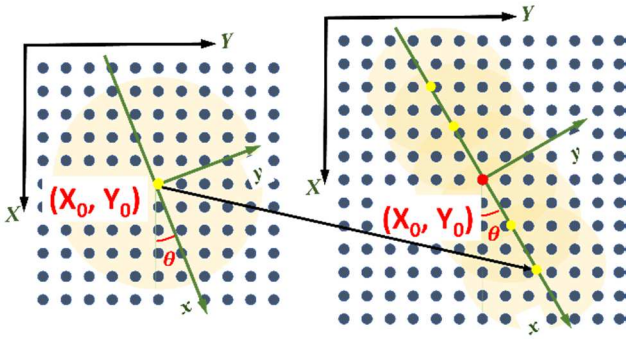

Figure 2: Line Filter Architecture

D. Line Filter Algorithm

The LF numerical algorithms and procedures need to be delicately formulated to improve computing efficiency and to generate a single set of LF operator coefficients (instead of iterative applications of multiple WVFs), which is extremely important for practical utilization. Additionally, the LF and WVF need to be applicable to parallelization to run efficiently on high-performance computing devices such as GPUs. The numerical procedure follows:

The half width of the LF (m) is defined, yielding the total length of the line, $2m + 1$ (also the number of pixels along the line), i.e., $2m + 1$ WV filters to construct the LF. The pixel neighborhood for each pixel along the line (including both the origin and the virtual pixels) is determined using the same approach as the WV filter. Their relative position from the origin are integers because the origin is a "real" pixel. This yields a set of neighbor pixels (used in the WV filter) for each virtual pixel, i.e., $\Delta_j = \{(x_1, y_1), (x_2, y_2), \dots (x_{N_p}, y_{N_p})\}_j$, where j is the index of the pixel along the line, and N_p is the number of neighbor pixels for the jth virtual pixel. To create a single set of the LF coefficients, we string together the neighbor pixels of all pixels along the line in the form of a union, i.e., $\omega = \{\Delta_1, \Delta_2, \cdots, \Delta_{2m+1}\}$. The total number of pixels in ω is N. This creates a single set that defines the range of the neighbor pixels for the entire LF, whose intensity values will be used for computing derivates of the LF. Then, the intensity values of all unique pixels surrounding the LF are retrieved by Equation (4).

$$\boldsymbol{F} = \begin{bmatrix} f(X_1, X_1) \\ \vdots \\ f(X_i, Y_i) \\ \vdots \\ f(X_N, Y_N) \end{bmatrix} \tag{4}$$

In Equation (4) $(X_i, Y_i) = (X_0, Y_0) + \omega(i)$ are the global coordinate of the i^{th}neighbor pixel in ω. The intensity values of neighbor pixels at the jth virtual pixel can be easily retrieved by $\boldsymbol{F}_j = \boldsymbol{P}_j \boldsymbol{F}$ where, $\boldsymbol{F}_j \in \boldsymbol{R}^{N_p}$ and $\boldsymbol{P}_j \in \boldsymbol{R}^{N_p * N}$ is a matrix to select the neighbor pixels around the jth virtual pixel out of all the neighbor pixels in the LF. $\boldsymbol{P}_j$ is a submatrix of the identity matrix. The numerical derivatives $\boldsymbol{z}_j$ are computed by the respective WV filter around the jth virtual pixel are given by Equation (5).

$$\boldsymbol{z}_j = \boldsymbol{A}_j^* \boldsymbol{F}_j = \boldsymbol{A}_j^* \boldsymbol{P}_j \boldsymbol{F} \tag{5}$$

In Equation (5) $\boldsymbol{A}_j^*$ represents the coefficients of the WVF at the jth pixels. Note that $\boldsymbol{A}_j^*$ is not constant for all pixels because the relative positions between each individual virtual pixel and its neighbor pixels vary along the line. Therefore, $\boldsymbol{A}_j^*$ depends on j. With all $\boldsymbol{A}_j^*$ available, the filter coefficients for the entire LF can be found using Equation (6).

$$\begin{aligned} \boldsymbol{z} &= w_1 \boldsymbol{z}_1 + \cdots w_j \boldsymbol{z}_j + \cdots w_{2m+1} \boldsymbol{z}_{2m+1} \\ &= \sum_{j=1}^{2m+1} w_j \boldsymbol{A}_j^* \boldsymbol{F}_j \\ &= \left(\sum_{j=1}^{2m+1} w_j \boldsymbol{A}_j^* \boldsymbol{P}_j \right) \boldsymbol{F} = \boldsymbol{A}_l \boldsymbol{F} \end{aligned} \tag{6}$$

$\boldsymbol{A}_l$ is the single set of coefficients for the entire LF and varies with the orientation of the LF and the number of the virtual pixels along the line.

IV. Results and Analysis

It is challenging to quantitatively evaluate the performance of gradient filters and edge detection as ground truth (GT) only comes from human labeled data or data produced by a subjective benchmark. For this reason, we first qualitatively analyzed the filters by performing gradient and edge comparisons to common and SOTA methods. Qualitatively, the most important evaluation criteria are gradient smoothness, edge gradient contrast, angular invariance, noise suppression, and edge connectivity. All these metrics can be assessed visually. A synthetic image was generated with varying signal-to-noise ratio and multiple lines at different angles, which provides a good baseline since all its parameters are known and adjustable. This test allows us to determine the limit of edge detection in the presence of noises. Afterwards the filters were applied to aquatic images with generally poor quality and compared with popular ML models. These results allow for a quantitative analysis of the methods when presented with real world scenarios containing hard to detect and small objects. Finally, numerical results were generated using a single scale Optimal Dataset Scale (ODS), Optimal Image Scale (OIS), and mean Aver Precision (mAP) score for the popular BIPED [16] and UDED [13] datasets. These metrics are standard for comparing edge detection techniques and are computed for our filters along with two SOTA ML methods DexiNed and TEED.

It is important to set fair standards when producing results for each method in comparison. Canny outlines a highly regarded process used to obtain edges from filter gradients [18] and as such, gradients from each operator are paired with the Canny determination process for obtaining edges in all results. Thresholds were statistically set at the same level, keeping all controllable variables constant.

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A. Synthetic Analysis

The first test is to judge the angular fidelity of the filters using the variably oriented lines of the synthetic image. Commonly used gradient filters like Sobel and Prewitt lack significant accuracy in determining diagonal line gradients since they estimate gradient magnitude using the global X and Y positions and only guarantee accurate detection along those directions [19]. Diagonal edges which do not align with the global X and Y axis will be approximations made from the two-direction decomposition. These yield weak direction responses represented by weak magnitudes. This issue is effectively tackled by our WVF and LF, which compute the gradients within a local coordinate system along a specific orientation, as shown in Figure 3. In this figure, each row corresponds to the gradients and edges for each synthetic image at a different SNR. Column (a) shows the original image, columns (b) and (c) display the gradients in the normal direction of LF when θ = 90° and 0°, and columns (d) through (g) compare the edges detected by our LF to those by BDCN and EDTER and DexiNed.

There are several interesting findings. First, when LF is at 90° (column (b) in Figure 3), it aligns with edges along the horizontal direction. This edge then shows the strongest presence of gradients in the normal direction of the filter (also the global X-axis). Conversely, the vertical edge exhibits a poor response as it is perpendicular to the LF. However, when the LF is oriented at 0°, i.e., column (c), all the observations are opposite. The vertical edge displays the strongest intensity of gradients along the normal direction of LF (also the global Y-axis), and the horizontal edge is almost non-detectable. In addition, edges in between the normal and tangential directions of LF will vary in intensity according to their angular distance from the tangential direction of LF. A closer angular distance creates stronger presence of its gradient value along the normal direction. The WVF and LF can be applied at any orientation and produce gradients at a very fine resolution.

In addition to angular sensitivity, the proposed WVF and LF excel in noise suppression. This is confirmed by results in columns (b) to (d) in Figure 3. The LF produces salient gradients and edges at an SNR of 1 (the first row). As the SNR decreases to SNR of 0.75, the edge can still be clearly detected, though its quality is slightly deteriorated. At an even lower SNR of 0.5, the edge map degrades but shows general geometry. The results demonstrate that the WVF and LF can detect edges at extreme SNR levels without using any smoothing filters, which protects the original photo information to preserve its integrity and use it at face value. Smoothing filters are necessary for other methods even at an SNR of 2 or higher (results not shown). It should also be considered that the filters can be expanded both in length and neighborhood to deal with increasing noise, or disconnected boundaries. The filter parameters can be adapted to the photo or used as a part of a scale space.

Last, our WVF and LF are compared against three highly regarded machine learning-based edge detectors, EDTER, BDCN, and DexiNed. The results show they cannot match the performance of the WVF and LF filters. They are heavily impacted by noise with unacceptable false positives and line breakages, and both completely miss detection when the SNR is less than 1. These models were not trained on data like

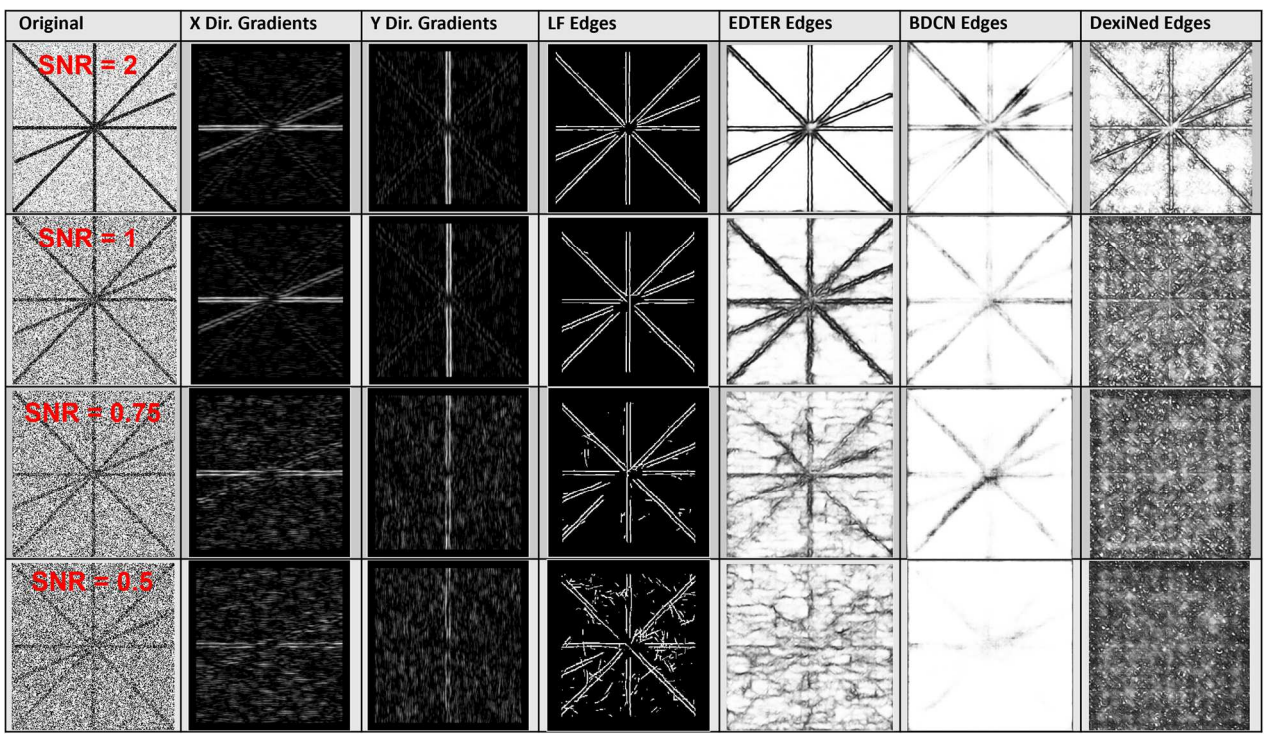

Figure 3: Gradients and Edges at Each Multi Line SNR Level Using Line Filter (Np=250 px and m=14px) and machine learning model Comparison (a) Original Photos; (b) Gradients @ θ = 90°; (c) Gradients @ θ = 0°; (d) LF Edges; (e) BDCN Edges; (f) EDTER Edges; and (g) DexiNed Edges

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synthetic images, which somewhat explains why they behave poorly in this case study. However, this highlights one of the concerns regarding deep learning models - a large volume of training data is needed to improve model generalization. In summary, the gradients and edges identified by LF show strong contrast to the noisy background, yielding edge maps which exhibit preferable contours. Although its results are not shown , WVF also consistently enhances edge connectivity and noise suppression. Some false positives and breakages do persist, which, however, can be effectively alleviated by the LF.

B. Aquatic Analysis

Since WVF and LF propose to calculate gradients at their precise orientation, they eliminate errors inherent in the directional decomposition of gradient based filters. Each filter also considers a larger number of neighboring pixels to suppress any noise in the calculation which has shown present issues for ML models like the results in Figure 3. The LF specifically chains the WVF together along a defined line to better calculate gradients in the presence of discontinuities. These aspects make the WVF and LF extremely useful for detecting smooth gradients of waves, objects floating in water, and any vessels, buoys, or debris which could be partially obscured while the variable filter architecture can be used to quickly set up a scale space capable of capturing edges on very small objects.

An example of superior gradient and edge detection is provided in Figure 4. This figure shows the gradients produced by the WVF and LF and compares them with two traditional methods Sobel and Prewitt. It particularly illustrates both filters' capabilities in suppressing noise, especially in the water region and highlights the line filter's ability to detect smooth uninterrupted line boundaries like those on the horizon. The buoys specifically are better captured by the WVF and LF. As expected, the WVF is slightly better than the LF in capturing the buoy edges because it is smaller and can fit sharp curves better while the LF is attuned for line detection and can sometime miss the areas of high curvature. This points to the power of the filters which complement each other by excelling in each other's weakness. The WVF can detect small objects with complex borders but is susceptible to noise while the LF is more capable of detecting lines even in the presence of noise and discontinuities but struggles with detailed shapes. The traditional methods which are compared in Figure 4 do not dominate active research which has favored learning methods for a while at present. Because of this, the comparison is extended to cover newer learning-based methods.

Since DexiNed is one of the top performing edge detection models with open-source code currently, it is used to compare against the WVF and LF in some aquatic photos which present challenging situations. The comparison which includes edge maps of various marine scenes and objects from the SOTA machine learning model DexiNed (Figure 5), shows better detection for the WVF and LF. In the first row, an underexposed image containing a boat and two buoys is presented. The challenge in this image is derived from the absence of light during exposure which creates low contrast boundaries on the objects. While all edge maps presented contain the leftmost buoy, the WVF and the LF are the only methods capable of capturing the boat and the rightmost buoy. In the middle image, a noisy yet straightforward buoy is presented and is fully captured by the WVF and LF. The other three methods, including DexiNed are only partially capable of accomplishing so. The final row presents a monochrome image of a braided cable which is nearly impossible to see without zooming in on the images. While the results for the WVF and LF are noisy, due to the water texture surrounding it, they do capture the cable whereas it is entirely uncaptured by Sobel, Prewitt, and DexiNed. The machine learning model is unable to produce any reasonable edges on the floating hose, likely

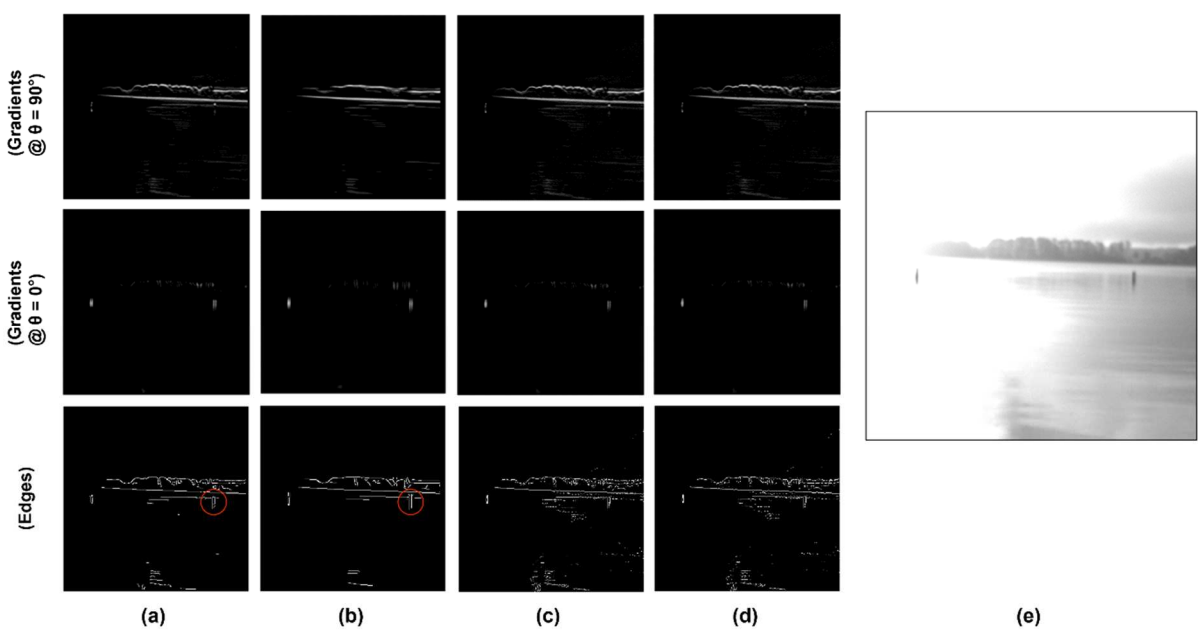

Figure 4: Gradient and Edge Comparison on Poor Quality Littoral Image (a) WVF (b) LF (c) Sobel (d) Prewitt

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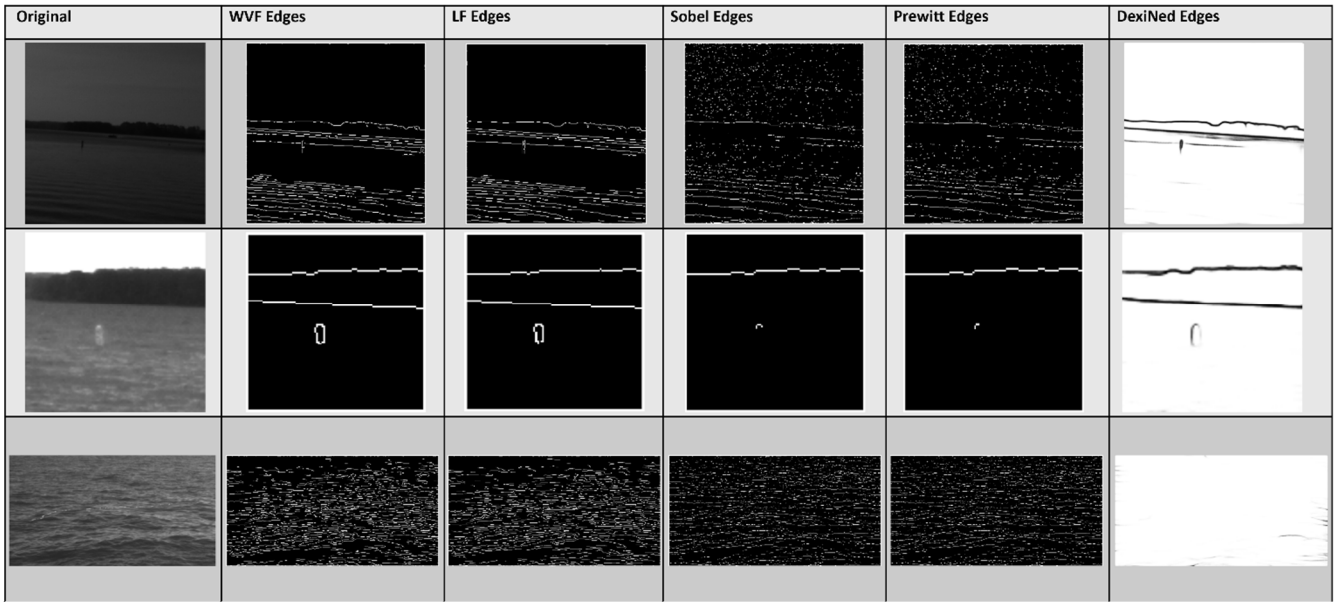

Figure 5: Edge Comparison on Various Aquatic Images Using the WVF, LF, Sobel, and ML Model DexiNed

because it has never seen an image of the sort. This highlights the problem of generalizability and data scarcity that deep learning models face.

C. Numerical Analysis

The final performance validation for the WVF and LF is a numerical analysis. The two filters were used to generate ODS, OIS, and mAP scores across two different datasets, UDED and BIPED. The ODS score measures the best f-score achieved across the dataset using a global threshold while the OIS score measures the average f-score across the dataset and mAP provides the average precision across the dataset. UDED is a carefully annotated dataset specifically designed for provided ground truth edge labels which compiles some of the more challenging images from many popular datasets. BIPED was also created specifically for generating ground truth data for edge detection and was generated from a consensus of human generated labels. BIPED is also the dataset that DexiNed and TEED were trained on, because it provides high quality labels.

The two filters are also compared against two SOTA models DexiNed, which is a larger model, and TEED which is a compact but accurate model. These have been chosen for their performance in edge detection and their difference in model size. Typically, when generating these results, edges are calculated using a scale space which accounts for error in ground truth labeling since edge locations are somewhat arbitrary because they are inherently discontinuities in the image. Additionally, they are hard to mark and inconsistent when generated by humans. However, since these filters are singular, it is only fair to test them on a single scale. Since they have displayed ideal behavior for generating an edge scale space, they can be tested in such a way down the road. The metrics presented in this analysis are therefore lower than normal but still valid, nonetheless.

The actual values generated for each edge detection method on each dataset can be found in Table 1. It can be seen that the WVF and LF outperform the learning-based methods in terms of ODS and OIS and are only slightly below TEED in mAP when looking at the results for the UDED dataset. This is highlighted by the green number in Table 1. When looking at the numbers for the BIPED dataset, the results are flipped. In this case, the ML models outperform our filters in terms of ODS and OIS but fall short in mAP. This could be a result of the fact that DexiNed and TEED were trained on BIPED and have seen similar data, however the test set is unfamiliar and should still be a useful evaluation.

This result highlights the benefit of the WVF and LF in that they do not require data to be trained on. These results we generated at face value which means the WVF and LF can produce comparable and even better edges while guaranteeing their results are generalizable and will not suffer from too specific or poor data. It is also likely these results can be improved with a scale space implementation.

Table 1: Numerical Analysis for WVF, LF, DexiNed, and TEED using BIPED and UDED Datasets

Method	EPOCH	ODS	OIS	mAP
DexiNed-UDED	11	.5541	.5516	.5587
DexiNed-BIPED	11	.6584	.6593	.5184
TEED-UDED	6	.6239	.6253	.6474
TEED-BIPED	6	.6976	.6980	.6528
WVF (Ours) – UDED	N/A	.6239	.6316	.6105
WVF (Ours) – BIPED	N/A	.6289	.6340	.6422
LF (Ours) – UDED	N/A	.6274	.6347	.5936
LF (Ours) – BIPED	N/A	.6326	.6406	.6621

Because we want to highlight the fact that the filters excel in challenging environments like the water, a small dataset of four images were labeled, each portraying a

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challenging water environment. These images were hand labeled to generate ground truth (GT) data and then tested in the same way to produce the same ODS, OIS, and mAP metrics. The images themselves and their corresponding labels can be seen in Figure 6. The images themselves are challenging because they have poor lighting conditions, lots of noise, high reflection, and small objects. Even with these challenges, the WVF and LF are able to generate impressive metrics. These can be seen in Table 2 which shows significantly higher OIS and mAP scores. DexiNed does deliver a slightly higher ODS score yet is much lower in the other metrics. On top of this, the actual edge probability maps generated by DexiNed and TEED do not capture many of the small objects in the images which would be of priority. So, while their numerical metrics may be higher in some cases, qualitatively, their edges are inferior.

Table 2: Numerical Analysis for WVF, LF, DexiNed, and TEED on Small Aquatic Dataset

Method	EPOCH	ODS	OIS	mAP
DexiNed - Aquatic	11	.6669	.6551	.5447
TEED - Aquatic	6	.5692	.6242	.5741
WVF (Ours) – Aquatic	N/A	.5960	.7038	.7358
LF (Ours) – Aquatic	N/A	.6173	.7212	.7043

While the metrics for TEED and DexiNed occasionally surpass those of our own filters, it is important to observe the true edge responses in conjunction. In the case of one of the aquatic images which contains two buoys, a boat, and a crab pot marker, both TEED and DexiNed are only capable of capturing a single buoy. All other objects are left unidentifiable. These cases are presented in Figure 7 and Figure 8 and the original photo is seen at the bottom left in Figure 6. Despite producing acceptable metrics, the images show how the actual results may not be acceptable in real operation. The results from the WVF show a much better response on the objects in frame, despite the poor exposure. These are shown in Figure 9 and the objects themselves are circled in red for enhanced visualization.

Figure 7: DexiNed Edge Prediction for Aquatic Image

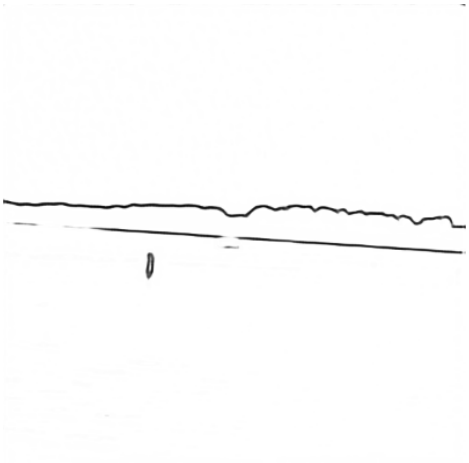

Figure 8: TEED Edge Prediction for Aquatic Image

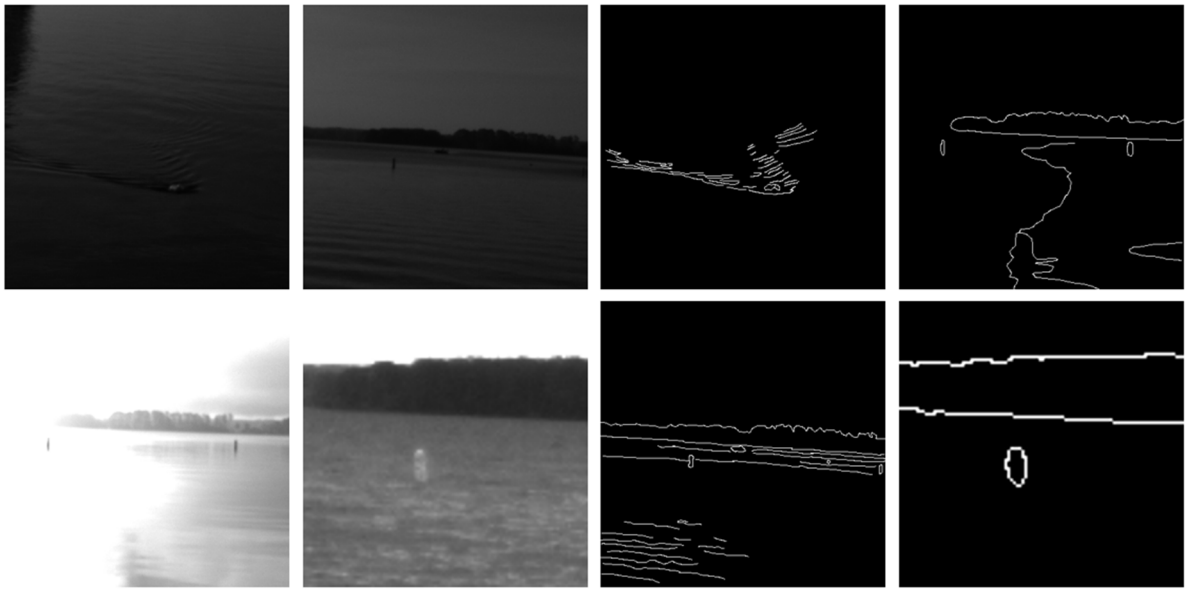

Figure 6: Small Aquatic Dataset and Associated Labels

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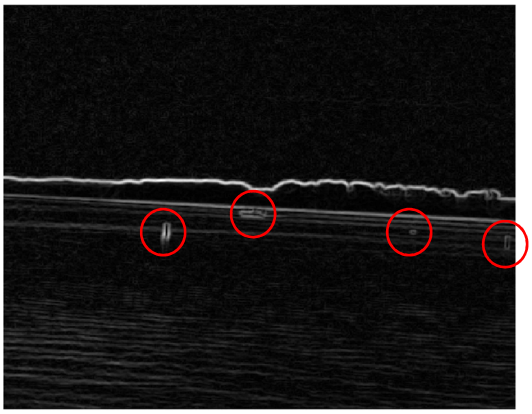

Figure 9: WVF Edge Prediction for Aquatic Image

V. Conclusion

Our numerical experiments confirm that the proposed filter architectures are capable of precise gradient calculation in environments with SNRs as low as 0.75 and provide exceptional foundations for downstream image processing techniques such as but not limited to edge detection. When compared with DexiNed, TEED, EDTER, and BDCN the edges produced by the WVF and LF are observationally smoother, better connected, and less phased by signal noise. They also numerically outperform the traditional operators and SOTA machine learning approaches by a salient margin for edge detection. In cases where the filters produce slightly lower metrics, their performance can still be justified by the detail of their edge maps. Many cases exist where important small objects are not captured by ML models but are by the WVF and LF. In addition, the WVF and LF have shown to be complementary in their performance and will be useful for generating accurate multiscale edges.

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